

RuFaS Economics Module Overview for the Financial Advisory Board (FAB)

Prepared for the April 25th FAB Meeting

Purpose of this Document: This document provides an overview of the approach and structure of the Ruminant Farm Systems (RuFaS) Economics Module, which integrates financial analysis into scenario-based decision support for dairy producers. The goal is to elicit feedback from dairy industry experts, CPAs, and farm financial advisors on the module's analytical approach, core assumptions, and key data inputs to ensure its relevance and practical utility before full deployment.

Module Purpose and Scope: The RuFaS Economics Module supports targeted evaluations of farm practices and technologies rather than whole-farm financial assessments. Its goal is to inform decisions around specific management changes or investments by quantifying tradeoffs in cost, revenue, and profitability.

Analytical Approach: At the core of the module is the **Flexible Economic Framework (FEF)**, which routes all economic assessments through a unified model that supports:

- **Partial Budget Analysis (PBA)** for evaluating changes in operating costs and revenues without capital investment.
- **Discounted Cash Flow Rate of Return (DCFROR)** analysis for long-term investment decisions, incorporating depreciation, financing, and cashflow timelines.

This consistent structure supports both single-scenario evaluations and comparative scenario analysis, with economic outputs including:

- **Internal Rate of Return (IRR):** The discount rate at which net present value (NPV) equals zero, representing the expected annual return on a capital investment.
- **Return on Investment (ROI):** The percentage ratio of net benefit to cost, used to measure how efficiently an investment generates profit through revenue gains or cost savings.
- **Payback Period:** The time it takes for an investment to recover its initial cost through net cash flows; shorter periods imply lower financial risk.
- **Profit Margin / Net Annual Cash Flow:** The difference between yearly revenues and operating costs, offering a snapshot of single-year profitability—especially useful for partial budget analyses.
- **Minimum Product Selling Price (MPSP):** The lowest price a product (e.g., milk) must earn to break even over the analysis period, showing sensitivity to market prices.

Supported Farm Practices: The module offers techno-economic evaluations for a range of dairy farm practices including:

- **Beef-on-Dairy Breeding Strategies:** Uses beef sires and often sexed semen to produce crossbred calves with higher market value than dairy bull calves. The module evaluates added semen costs, calf revenue, and the balance between replacement needs and profitability.
- **High-Quality Feed for Increased Milk Production:** Assesses whether upgrading feed with more digestible or nutrient-rich ingredients boosts milk output enough to justify higher feed costs under varying milk prices and productivity levels.

- **Methane-Reducing Feed Strategies:** Evaluates feed adjustments (e.g., fats, tannins, additives) that cut enteric methane emissions and potentially earn carbon credits, factoring in both cost and effectiveness.
- **Feed Quality and Manure Nutrient Value:** Considers how improved feed quality raises manure nutrient content, which can reduce fertilizer needs. Quantifies the tradeoff between feed cost and potential fertilizer savings.
- **Land Allocation for Feed Cost Reduction:** Examines shifting acreage to homegrown feed (e.g., corn silage, alfalfa) to reduce purchased feed costs and improve economic performance while maintaining feed quality.
- **Conservation Cropping Practices:** Covers no-till, crop rotation, and cover cropping, which enhance soil health, reduce input costs, and improve yield stability. Also assesses potential soil carbon credit revenue alongside implementation costs.
- **Manure Management System Investments:** Analyzes systems like digesters, separators, and storage upgrades for their ability to reduce nutrient loss, lower fertilizer and hauling costs, and generate revenue from energy and carbon credits.
- **Animal Comfort Improvements:** Looks at barns, bedding, and ventilation upgrades that improve herd health and milk yield while lowering labor and veterinary costs. Evaluates returns relative to investment.
- **Feed Storage Infrastructure Enhancements:** Assesses upgrades that reduce feed shrink and spoilage, preserving nutritional value and lowering feed costs. Evaluates potential savings and improved ration consistency.

Capital and Operational Costs: The economics module quantifies both capital expenditures (CAPEX) and operational costs (OPEX) by linking directly to outputs from the RuFaS biophysical module, supplementing with user inputs or default values from the RuFaS Economic Database where biophysical data are unavailable or slated for inclusion in future model iterations. Biophysical outputs—such as manure volume, fuel consumption, energy use, and nutrient balances—are multiplied by commodity prices and service rates to estimate economic impacts.

The model captures a broad range of operational costs and product revenues by combining biophysical outputs with commodity prices from the RuFaS Economic Database including:

- **Feed Ingredients:** Common feed ingredients (with default data for all regional baseline rations)
- **Energy:** Electricity, natural gas, propane, diesel, and gasoline, with prices differentiated by sector (e.g., residential, commercial, industrial)
- **Water:** Regionalized costs for both irrigation and municipal water use
- **Fertilizers:** Prices for common fertilizers
- **Labor:** Hired labor costs, using regionally resolved wages from USDA datasets across the U.S.
- **Farm Services:** Veterinary care, manure disposal, land rental, tractor operation, and equipment maintenance
- **Additional Consumables:** Bedding materials, semen, crop seeds, and RNG upgrading consumables
- **Products:** Milk (with price data by milk class), livestock (heifers, steers, calves by class and breed), carbon credits (enteric, energy production, soil carbon), manure sales, renewable natural gas (RNG) and electricity sales, and other co-products like bedding and digestate (fertilizer).

Capital costs are calculated using scale-sensitive functions for equipment such as digesters, solids separators, RNG systems, and combined heat and power (CHP) units. Each subsystem includes installation multipliers and allows for either user-defined or default inputs, with dynamic cost scaling based on throughput or herd size.

By combining these system-level outputs and cost databases, the economics module enables flexible, robust evaluation of farm investments and operational strategies.

Economic Inputs and Default Values: Economic parameters are sourced from:

1. **User inputs** (highest priority)
2. **County-level or regional data** (if available)
3. **National averages** (fallback)

When historical economic data is used—particularly in DCFROR analyses—the Inputs Manager adjusts for inflation by indexing all values to a common dollar year. All historical values in the economic database are reported in nominal terms for the year they were observed (e.g., 2020 dollars for data from 2020). To ensure compatibility with DCFROR logic, the Inputs Manager deflates or inflates these values to match the starting year of the simulation. For example, if the simulation spans 2020–2024, all economic inputs will be adjusted to 2020 dollars before applying discounting using Consumer Price Index data from the United States Bureau of Labor and Statistics.

Users conducting a price volatility analysis can also provide a vector of input prices across years. In this case, the Inputs Manager assumes the user has already adjusted prices to the appropriate dollar year and will require a flag indicating whether all values are in the same currency year. Failure to standardize dollar years across inputs may result in inconsistent or misleading DCFROR results, so users must ensure consistency before submitting vectors of price data to evaluate market volatility.

Data Gaps and Advisory Needs The project team is seeking FAB guidance to close the following data gaps:

1. **Bedding Material Costs:** Regional prices for sand, straw, sawdust, and manure solids.
2. **Semen Prices and Usage:** Current cost trends and use of beef vs. sexed semen across regions.
3. **Carbon Credits:** Relevance, valuation, and markets for carbon credits
4. **Crossbred vs. Dairy Calf Pricing:** Regional prices for Holstein vs. Holstein/beef crossbred calves and cows. While USDA national averages are available, recent spikes in calf value need validation for both dairy and crossbred beef markets. We are also seeking FAB input on whether default values for animal/calf pricing are appropriate or whether they are too variable across time, region, and market type to standardize.

Next Steps The RuFaS team will present the current model structure and walk through examples during the April 25th board meeting. Your feedback will shape revisions to both the module logic and the underlying economic dataset.

Thank you for your partnership and expertise!